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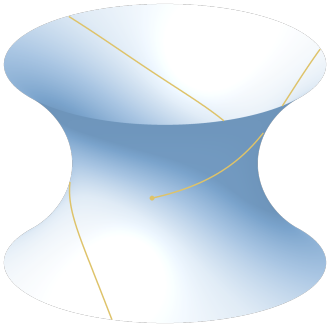
Federal Fraud Expansion
Working Document

Forensic methods handbook

The mathematics behind each fraud test

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$$\mathcal{G}(x) = x^x \qquad \nabla_{\dot\gamma}\dot\gamma = 0$$

We navigate our clients from a challenging problem to a defensible solution along the shortest rigorous path.

Contents

Bottom line **2**

1. Where each method sits **2**

2. Ground rules for forensic testing **3**

3. Digit and number tests **3**

3.1 Benford's law . . . 3

3.2 Companion number tests . . . 4

4. Duplicate, split and impossible-billing tests **5**

4.1 Duplicate payments . . . 5

4.2 Split purchases and split awards . . . 5

4.3 Impossible days and impossible volumes . . . 6

4.4 Services and payments after death . . . 6

5. Peer outliers, growth and threshold bunching **6**

5.1 Peer groups . . . 6

5.2 Robust z-scores . . . 6

5.3 Count data and empirical Bayes shrinkage . . . 7

5.4 Bunching below thresholds . . . 7

5.5 Rapid growth and change points . . . 7

5.6 Concentration . . . 7

6. Collusion screens for sealed bids **8**

7. Entity resolution **8**

8. Network analysis **9**

8.1 Shared-attribute graph . . . 9

8.2 Co-location test . . . 9

8.3 Components, brokers and communities . . . 9

8.4 Risk propagation . . . 10

9. Eligibility and identity tests **10**

10. Estimating improper payments **10**

10.1 Stratified expansion estimator 11

10.2 Ratio estimator 11

10.3 When the normal interval fails 11

10.4 Monetary unit sampling 11

10.5 Conventions to know 12

10.6 Checking a published estimate 12

11. Reconciliation and audit-trail tests **12**

11.1 Public reconciliations (Phase 1) 12

11.2 Audit-trail tests on internal ledgers (Phase 2) 13

12. Combining signals, calibrating and validating **13**

12.1 Likelihood ratios 14

12.2 The base-rate trap 14

12.3 Validation metrics 14

12.4 Controlling false discoveries 14

13. Methods by fraud area **15**

14. Reference implementation **16**

Sources **17**



Status: Draft v0.1 (2026-11-03), for Steve's review. Companion to the methodology, which says how a lead is scored, validated and referred; this handbook says exactly how each test is computed, when it is valid and what it cannot show. Every formula here has a tested reference implementation in `analytics/forensic/` (run on synthetic data only; no real entity appears in the tests).

Bottom line

Fraud leaves arithmetic traces: digits that people invent, amounts steered just under a limit, invoices paid twice, bids that cluster too neatly, providers who bill more hours than a day has, and "separate" companies that share a phone line. This handbook turns each trace into a written test with a formula, a threshold, a false-positive story and a validation plan. Ten families of tests cover every fraud area in the roadmap and in the all-agency plan:

1. **Digit and number tests** (Benford's law, round numbers, duplication).
2. **Duplicate, split and impossible-billing tests.**
3. **Peer outliers, growth and threshold bunching**, with Bayesian shrinkage so small providers are not over-flagged.
4. **Collusion screens** for sealed bids.
5. **Entity resolution** (Fellegi-Sunter record linkage).
6. **Network analysis** with hub discounting and risk propagation.
7. **Eligibility and identity tests** (death, incarceration, impossible identities).
8. **Improper payment estimation** with confidence intervals, so we can check and extend agency estimates.
9. **Reconciliation and audit-trail tests** across USAspending, Treasury, PaymentAccuracy.gov and Single Audit data.
10. **Score combination and validation** (likelihood ratios, lift, false discovery control).

The rule that keeps this honest: **a test result is a reason to look, not a finding.** Only an Inspector General or DOJ confirms fraud, and only a confirmed dollar counts as savings.

1. Where each method sits

Tests work at five levels. Public data supports the first three in Phase 1; the last two mostly need internal records reached through OIGs in Phase 2.

Level	Unit tested	Example tests	Phase 1 public data	Phase 2 internal data
Populatio	All awards or payments of an agency or program	Benford, round numbers, bunching, reconciliation	USAspending award and account files, PaymentAccuracy.gov, Treasury MTS	General ledger, payment files
Entity	One contractor, grantee, provider, retailer	Peer outliers, growth, new-entity-large-award, entity-level Benford	USAspending, SAM.gov, CMS files, SNAP retailer data	Claims, invoices



Level	Unit tested	Example tests	Phase 1 public data	Phase 2 internal data
Network	Groups of entities	Shared attributes, components, risk propagation	SAM.gov addresses and contacts, NPPES, Form 990, state registries	Bank accounts, devices, IP addresses
Transacti	One payment, claim, invoice or voucher	Duplicates, splits, impossible days, after-death payments	Rarely public	Payment, claim, card and voucher files
Estimate	The program's improper payment rate	Stratified and ratio estimators, monetary unit sampling	Published agency estimates	Sample case files

2. Ground rules for forensic testing

- **Hypothesis first.** Each test is written down with the scheme it targets before it is run. A test without a scheme is data dredging.
- **Pre-registration.** Thresholds, peer groups and exclusions are fixed in the screen's specification before results are seen, and changes are logged with reasons.
- **Multiple testing.** Running one test per entity across 100,000 entities guarantees thousands of chance "hits". Entity-level p-values are controlled with the Benjamini-Hochberg procedure (section 12.4), not a fixed 0.05.
- **Practical significance.** With millions of records, a chi-square test rejects for trivially small deviations. Every test reports an effect size (MAD, ratio, excess dollars) alongside its p-value, and effect size decides whether anyone looks.
- **Innocent explanations first.** Each test lists its known benign causes and checks the main ones before a result enters a lead package.
- **Reproducibility.** Every result cites its data pull (source, date, hash) and the code version, so another analyst can rerun it exactly. This follows the GAO Yellow Book's evidence standard and the AICPA Statement on Standards for Forensic Services No. 1.
- **Known error rates.** Each screen's back-test (section 12.3) gives its measured hit rate and false-positive rate. That is what an expert needs to defend a method in court (the Daubert standard asks for a known or potential error rate).

3. Digit and number tests

3.1 Benford's law

Many naturally occurring sets of amounts (invoices, claims, award values) follow Benford's law: the leading digit d appears with probability

$$P(D_1 = d) = \log_{10}\left(1 + \frac{1}{d}\right), \quad d = 1, \ldots, 9,$$

so about 30.1% of amounts start with 1 and only 4.6% start with 9. The first-two-digits version, with $d = 10, \ldots, 99$, uses the same formula and is the sharper forensic test. The second digit has probability



$$P(D_2 = d) = \sum_{k=1}^{9} \log_{10}\left(1 + \frac{1}{10k + d}\right), \quad d = 0, \ldots, 9.$$

Why it detects fraud. People inventing numbers spread leading digits too evenly and favor middle digits; schemes that keep amounts under a limit create spikes at specific first-two digits (for example 48 and 49 below a $5,000 approval level).

When it applies. Benford's law holds for amounts that span several orders of magnitude, arise from multiplying many factors, and have no built-in floor or ceiling. It does **not** apply to assigned numbers (invoice or account numbers), prices set on a schedule (fixed per-diem rates, Medicare fee schedules), amounts with caps (a $10,000 grant maximum), or small samples. A thousand or more records works best for first-two-digit tests.

Statistics. With n amounts, observed digit shares $\hat{p}_d$ and expected shares p_d:

- Chi-square: $\chi^2 = n \sum_d (\hat{p}_d - p_d)^2 / p_d$ with 8 (first digit) or 89 (first two) degrees of freedom. Sensitive to n; use for small populations.
- Per-digit z with continuity correction: $z_d = \left(|\hat{p}_d - p_d| - \frac{1}{2n}\right) / \sqrt{p_d(1 - p_d)/n}$. Identifies which digits spike.
- Mean absolute deviation: $\text{MAD} = \frac{1}{k} \sum_d |\hat{p}_d - p_d|$, which does not grow with n and is the main conformity measure.

Test	Close conformity	Acceptable	Marginal	Nonconformity
First digit	MAD up to 0.006	0.006 to 0.012	0.012 to 0.015	Above 0.015
Second digit	Up to 0.008	0.008 to 0.010	0.010 to 0.012	Above 0.012
First two digits	Up to 0.0012	0.0012 to 0.0018	0.0018 to 0.0022	Above 0.0022

Thresholds are Nigrini's (2012), derived from large real-world data sets.

How we use it. At the population level, nonconformity is a reason to look at the program's controls, not at any one payee. At the entity level (one vendor's 400 invoices, one provider's claims), each entity's MAD is compared with its peers' distribution of MADs, and only the extreme tail with enough records is reviewed. Benford is a **weak signal alone**; it never counts as one of the two required signals unless combined with a digit spike at a known threshold.

3.2 Companion number tests

Test	Formula	What it catches	Benign causes
Last-two-digits	Chi-square of the last two digits against uniform (1% each)	Invented cents, or amounts all ending .00 or .99	Whole-dollar pricing, fixed fees
Round numbers	Share of amounts that are exact multiples of $1,000 (or $100), tested against a clean reference share with a one-sided binomial test	Estimates billed as actuals; made-up invoices	Lump-sum milestone payments, fixed grants



Test	Formula	What it catches	Benign causes
Number duplication	Most frequent exact amounts, ranked	One odd amount (say $4,987.13) paid 60 times	Recurring contracts, subscriptions
Summation	Share of total dollars by first-two digits; each of the 90 groups should hold about 1/90	A few very large amounts driving the total	One large legitimate contract
Relative size factor	Largest payment divided by second largest, per vendor or cardholder	One inflated or misplaced payment	Annual true-ups

4. Duplicate, split and impossible-billing tests

These are transaction tests. They need payment, invoice, card or voucher records, so most run in Phase 2, but the award-level versions run in Phase 1 on USAspending contract actions.

4.1 Duplicate payments

Duplicate payments are the most common recoverable error and a common fraud (resubmitting an invoice with a changed number). The test runs in four tiers, strongest first:

1. **Exact:** same vendor, amount, invoice number and date.
2. **Same invoice, different date:** same vendor, amount and *normalized* invoice number. Normalization uppercases, strips prefixes (INV, #) and suffixes (-A, -R, DUP), maps look-alike characters (O to 0, I and l to 1, S to 5) and removes leading zeros.
3. **Near-identical invoice:** same vendor and amount, normalized invoice numbers within one edit (Damerau-Levenshtein distance 1, which counts a transposed pair of digits as one edit), paid within 90 days.
4. **Same invoice, different vendor:** same amount and invoice number paid to two vendor records, which points to duplicate vendor master entries or a diverted payment.

Expected false positives are recurring identical charges (monthly rent, subscriptions); these are listed per vendor and excluded after review.

4.2 Split purchases and split awards

Rules that require more scrutiny above a dollar limit (the FAR micro-purchase threshold for card purchases, the simplified acquisition threshold for contracts, internal approval levels) invite splitting. The test groups purchases by the same buyer from the same vendor within a short window (three days for cards, 30 days for contract actions) and flags any group where **every** item is under the limit but the **sum** is over it.

At the population level, the same idea becomes a **bunching test** (section 5.4): an excess of amounts just under the limit.

The limits change with inflation adjustments (the October 2025 FAR adjustment raised the micro-purchase threshold to $15,000 and the simplified acquisition threshold to $350,000); each screen reads the limit in force on the transaction date from FAR 2.101 rather than a fixed number.



4.3 Impossible days and impossible volumes

For any biller paid by time or by service (Criminal Justice Act panel attorneys, workers' compensation providers, Medicaid personal care aides, time-and-materials contractors), total billed hours per person per day across all matters is

$$H_{b,t} = \sum_{m} h_{b,m,t}.$$

$H > 24$ is impossible; reviews also look at $H > 16$ on many days and at service counts that exceed what one clinician can deliver (for example, more 60-minute psychotherapy sessions in a day than hours in a working day). The test is simple, explains itself to a jury, and has produced many health care fraud prosecutions.

4.4 Services and payments after death

Any service or payment dated after the beneficiary's date of death, beyond a short grace period for legitimate end-of-month payments, is flagged. Death data come from SSA through Treasury's Do Not Pay service (which began receiving SSA's full death file in 2023 under a three-year pilot) or from state vital records; the Phase 1 public alternative is the obituary-backed case library, which is too sparse to screen with.

5. Peer outliers, growth and threshold bunching

5.1 Peer groups

Every outlier test compares an entity with **peers in the same program, service, place and period**: a hospice with hospices in its county and year, a SNAP retailer with stores of the same type within a mile. A national comparison mostly finds specialty, case mix and size.

5.2 Robust z-scores

For a peer group with values x_i,

$$z_i = \frac{x_i - \text{median}(x)}{1.4826 \cdot \text{MAD}(x)},$$

where $\text{MAD}(x) = \text{median}\,|x_i - \text{median}(x)|$. The constant makes the scale match a standard deviation for normal data, and unlike the mean and standard deviation, a few extreme entities cannot mask themselves by inflating the scale. Skewed measures (dollars, counts) are log-transformed first. Review threshold: $z > 3.5$ (set per screen in the calibration week).



5.3 Count data and empirical Bayes shrinkage

For counts (claims, services, meals) with exposure (beneficiaries, enrolled children), let entity i have observed count y_i and expected count E_i from its peers' rate. The raw ratio y_i/E_i is unstable when E_i is small: 3 claims against 0.5 expected looks like a 6x outlier but is weak evidence. We model

$$y_i \sim \text{Poisson}(\theta_i E_i), \qquad \theta_i \sim \text{Gamma}(\alpha, \beta),$$

fit α, β across the peer population by the method of moments, and rank on the posterior

$$\theta_i \mid y_i \sim \text{Gamma}(\alpha + y_i,\ \beta + E_i), \qquad \mathbb{E}[\theta_i \mid y_i] = \frac{\alpha + y_i}{\beta + E_i}.$$

Leads are ranked by the posterior probability that θ_i exceeds a material multiple of peers (1.5x by default), which balances size of excess against strength of evidence. This is the same approach used in disease mapping and in the hierarchical models behind CMS's hospital readmission and mortality measures.

5.4 Bunching below thresholds

If a limit T invites steering, amounts pile up just below it. Bin amounts in width w, fit a polynomial of degree 5 to bin counts outside a window around T, and predict counts inside the window. Excess mass is

$$B = \sum_{\text{bins below } T \text{ in window}} (c_j - \hat{c}_j), \qquad b = B \Big/ \sum \hat{c}_j,$$

following Chetty et al. (2011). A ratio b well above zero, with matching missing mass just above T, is strong population-level evidence of splitting or steering. Applied to card purchases (micro-purchase limit), contract actions (simplified acquisition and certified cost data thresholds) and grant draws (prior-approval thresholds). Standard errors come from bootstrapping the residuals.

5.5 Rapid growth and change points

For an entity's monthly payments x_t with a peer-based expected level μ_0, the one-sided CUSUM

$$S_t = \max(0,\ S_{t-1} + x_t - \mu_0 - k)$$

signals when $S_t > h$. Setting k to half the shift worth detecting and h at four to five standard deviations gives a low false-alarm rate. CUSUM catches sudden ramps (a new hospice billing at the cap within months, a retailer whose redemptions triple) that annual totals hide.

5.6 Concentration

The Gini coefficient of an entity's billing across service codes, or of a program's payments across entities, measures concentration. A provider whose billing collapses onto one or two high-paying



codes (Gini near 1 where peers sit near 0.5) is an upcoding signal; a program where a handful of new entities take most of the money is a capture signal.

6. Collusion screens for sealed bids

Bid rigging leaves statistical traces in the full set of bids, which state DOTs publish as bid tabulations (DOT-1) and some agencies release on request. For each tender with bids $b_{(1)} \leq b_{(2)} \leq \cdots \leq b_{(n)}$:

Screen	Formula	Collusive pattern
Coefficient of variation	$\mathrm{CV} = \sigma_b / \bar{b}$	Low: cover bids cluster near an agreed price
Spread	$(b_{(n)} - b_{(1)})/b_{(1)}$	Low
Relative distance	$\mathrm{RD} = (b_{(2)} - b_{(1)})/\sigma_{\text{losing}}$	High: a gap between the designated winner and tight cover bids
Skewness and kurtosis	Third and fourth standardized moments of the bids	Distorted by clustered cover bids
Cover ratio	Mean losing bid divided by winning bid	Stable across tenders for the same group

Imhof, Karagök and Rutz (2018) found these distribution screens separate known Swiss cartels from competitive markets, and Huber and Imhof (2019) raised accuracy by combining them in a classifier. Our use: compute the screens per tender, then compare each regular group of bidders with the same work type and region.

Rotation test. Among a stable group of firms, cartels often take turns. Count how often the same firm wins two consecutive tenders, and compare with a permutation distribution formed by shuffling the order of the same wins. Too *few* repeats (an orderly rotation) is the signal.

Market allocation. A win-share matrix shows, for each pair of firms that often bid together, who wins. Firms that meet often but each win only in "their" county or season are flagged for review.

Benign causes. Few bidders in rural markets, standard engineer's estimates that anchor bids, capacity constraints. Screens are leads for DOJ's Procurement Collusion Strike Force, not findings.

7. Entity resolution

The entity spine joins records that describe the same business or person across SAM.gov, USAspending, NPPES, Form 990, state registries and the watch lists. Errors here create false networks (merging strangers) or hide real ones (splitting one entity).

Blocking. Compare only pairs that share a cheap key (ZIP plus name token, phone, email domain, UEI, NPI), which reduces billions of pairs to millions.

Comparison. For each candidate pair, compare fields to get an agreement vector γ: name (Jaro-Winkler similarity at least 0.92 after removing legal suffixes), street address (USPS-standardized, exact), phone (exact), officer name, email domain.

Fellegi-Sunter weights. With $m_k = P(\gamma_k = 1 \mid \text{match})$ and $u_k = P(\gamma_k = 1 \mid \text{non-match})$, the match weight is



$$W = \sum_k \left[\gamma_k \log_2 \frac{m_k}{u_k} + (1 - \gamma_k) \log_2 \frac{1 - m_k}{1 - u_k} \right].$$

A rare field agreement (same phone number) earns a large weight because u_k is tiny; a common one (same city) earns little. m and u are estimated without labels by the EM algorithm and checked on the 500 hand-labeled pairs in the methodology.

Decision. Pairs above an upper threshold are matches; below a lower threshold, non-matches; between them, clerical review. The upper threshold is the lowest weight with labeled precision of 95% or better.

Clustering. Linking matched pairs transitively can chain unrelated records (A matches B, B matches C, A and C are strangers). We cluster on the weighted match graph and split clusters whose weakest internal link falls below the upper threshold.

8. Network analysis

8.1 Shared-attribute graph

Entities connect through shared attributes: address, phone, email, officer, registered agent and, in Phase 2, bank account and device. We build the bipartite entity-attribute graph and project it onto entities with an Adamic-Adar style weight:

$$w_{ij} = \sum_{a \in A_i \cap A_j} \frac{1}{\log(1 + \deg(a))}.$$

A phone shared by two firms counts far more than an address shared by 4,000 firms at a registered agent. Attributes on the benign-hub list, or shared by more entities than a set cap, are dropped.

8.2 Co-location test

Is an address hosting more entities than its type explains? If addresses of that type (single-family home, small office, mailbox store) average λ registered entities, the Poisson tail

$$P(X \geq k) = 1 - \sum_{i=0}^{k-1} \frac{e^{-\lambda} \lambda^i}{i!}$$

gives how surprising k entities are. Twelve federal contractors at a house where 1.5 are typical is a lead; twelve at a coworking space is not.

8.3 Components, brokers and communities

- **Connected components** above an edge-weight floor give candidate rings.
- **Betweenness centrality** finds brokers (one consultant, preparer or billing agent linking many outliers), a pattern in E-Rate, SBA lending and health care cases.



- **Community detection** (Louvain or Leiden) splits large components into tighter groups for review.
- **Temporal motifs** catch waves: many entities registered within weeks of each other at linked addresses, just before a program opens.

8.4 Risk propagation

Personalized PageRank spreads risk from known bad actors (excluded, debarred, convicted entities) through weighted links. With restart probability α and seed distribution s,

$$r = \alpha s + (1 - \alpha) P^{\top} r,$$

where P is the row-normalized weighted adjacency matrix. An entity's score measures how close it sits to known fraud through strong links. Used as a signal, never as guilt by association: a high score must be paired with an independent signal of its own.

9. Eligibility and identity tests

Mostly Phase 2, run on applicant and payee files under OIG data agreements.

Test	Rule	Programs
Death match	Payment or service date after date of death (SSA full death file, state records)	Medicare, Medicaid, FEHB, VA, PBGC, RRB, SNAP, OPM annuities
Incarceration match	Benefit paid during a period of incarceration (BOP and state prison data)	SSI, UI, SNAP, VA
Impossible identity	SSN issued after the person's stated birth year plus a plausible gap, or SSN issued after the claim; SSNs issued after June 2011 are randomized and need SSA verification instead of area-number rules	UI, SBA, student aid
Identity reuse	One SSN, bank account, phone, email or device behind claims in several states or programs	UI, SNAP, Medicaid, SBA, ED
Synthetic identity	Thin credit-file-style profile: new email, prepaid phone, no history, shared device	UI, SBA, tax credits
Concurrent eligibility	Benefits that cannot legally be received together (for example some disability and earnings combinations)	SSA, RRB, DOL, VA

Bank-account reuse is the strongest single identity signal in pandemic cases; it is the core of cross-agency idea X-2 in the roadmap.

10. Estimating improper payments

Agencies estimate improper payments by sampling payments and reviewing them (Payment Integrity Information Act of 2019; OMB Circular A-123, Appendix C). We need this math for



three jobs: reading published estimates critically, designing samples for Phase 2 pilots, and estimating the dollars behind a screen's leads without overstating them.

10.1 Stratified expansion estimator

Payments are split into strata h (by size, region or payment type). From N_h payments, n_h are sampled and reviewed, giving improper dollars y_{hi}. The estimated total and its variance are

$$\hat{Y} = \sum_h N_h \bar{y}_h, \qquad \widehat{\mathrm{Var}}(\hat{Y}) = \sum_h N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_h^2}{n_h}.$$

The improper payment rate is $\hat{R} = \hat{Y}/X$, where X is total outlays, with standard error $\mathrm{SE}(\hat{Y})/X$. A two-sided interval at confidence $1 - a$ is $\hat{Y} \pm z_{1-a/2}\,\mathrm{SE}$.

10.2 Ratio estimator

When improper dollars scale with payment size, the combined ratio estimator is more precise:

$$\hat{R} = \frac{\sum_h N_h \bar{y}_h}{\sum_h N_h \bar{x}_h}, \qquad \widehat{\mathrm{Var}}(\hat{R}) = \frac{1}{\hat{X}^2} \sum_h N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_{d,h}^2}{n_h},$$

where x is dollars paid, $\hat{X} = \sum_h N_h \bar{x}_h$ and $d = y - \hat{R}x$ are the residuals (Taylor linearization).

10.3 When the normal interval fails

Improper dollars are highly skewed: most sampled payments are fully correct, a few are wholly wrong. Normal-theory intervals then cover the truth less often than stated. In our simulation (8% of payments improper, lognormal payment sizes; see `analytics/tests/test_forensic.py`), nominal 90% intervals covered the true rate about 77% of the time at $n = 100$, 84% at $n = 400$ and 88% at $n = 1{,}500$. Remedies, in order of preference: larger samples in high-variance strata (Neyman allocation, $n_h \propto N_h S_h$), bootstrap intervals, and for audits with few errors, monetary unit sampling.

10.4 Monetary unit sampling

Monetary unit (dollar unit) sampling selects dollars, not payments, so large payments are more likely to be sampled. With book value BV, n sampled dollars and taintings $t_{(1)} \geq t_{(2)} \geq \ldots$ (each error's share of its payment's value), the Stringer upper error limit is

$$\mathrm{UEL} = \frac{BV}{n} \left[U(0) + \sum_{i \geq 1} \big(U(i) - U(i-1)\big)\, t_{(i)} \right],$$



where $U(k) = \frac{1}{2}\chi^2_c(2k+2)$ is the upper Poisson confidence limit after k errors ($U(0) = 3.00$ and $U(1) = 4.74$ at 95%). The sample size that keeps the bound under a tolerable misstatement TM when no errors are expected is $n = BV \cdot U(0)/TM$.

10.5 Conventions to know

- HHS-OIG and Medicare contractors extrapolate overpayments using the lower limit of a confidence interval (HHS-OIG uses the lower limit of a two-sided 90% interval), which favors the provider. Our lead packages use the same convention when they size dollars.
- HHS-OIG's RAT-STATS is the standard tool for these samples; our code reproduces its core estimators for checking.
- **Unknown payments** (payments that cannot be shown proper for lack of documentation) are reported separately from improper payments under A-123 and are neither fraud nor savings.
- **Improper is not fraud.** GAO estimated federal fraud losses at $233 billion to $521 billion a year (GAO-24-105833) using a different method; FY2025 improper payment estimates totaled about $186 billion across 64 programs at 15 agencies (GAO-26-108694). Neither figure is ever used as a savings claim.

10.6 Checking a published estimate

From public reports we can test whether an estimate is internally consistent: whether its stated interval matches its sample size and strata, whether the rate moved more than its standard error would allow from one year to the next without a stated cause, and whether programs that "did not report" left large susceptible outlays unmeasured. These checks become questions in a Phase 2 proposal, not findings.

11. Reconciliation and audit-trail tests

11.1 Public reconciliations (Phase 1)

Check	Left side	Right side	What a gap means
Account to award	USAspending File A/B obligations and outlays by Treasury account	File C award-level obligations linked to D1/D2 awards	Unlinked or unreported awards; spending invisible to payee-level screens
Agency totals	USAspending agency outlays	Treasury Monthly Treasury Statement outlays	Reporting scope differences (trust funds, offsetting receipts); explains, rarely fraud
Program outlays	PaymentAccuracy.gov susceptible outlays by program	USAspending outlays by Assistance Listing	Programs measured on a different base than they pay out
Prime to subaward	Prime award obligation	Sum of FFATA subawards reported under it	Subawards exceeding the prime are impossible and point to reporting errors or double-counting; missing subawards hide pass-throughs



Check	Left side	Right side	What a gap means
Grantee self-report	Grant drawdowns in USAspending	Federal expenditures reported in the grantee's Single Audit	Large unexplained gaps between what was drawn and what was audited
Single Audit findings	Federal Audit Clearinghouse findings and questioned costs	The grantee's continuing awards	Repeat material weaknesses or questioned costs with no change in funding

The Single Audit data (fac.gov) are the broadest public internal-control data in government: every non-federal entity spending $1 million or more of federal awards a year (the threshold since the 2024 Uniform Guidance revision) files one. Two cheap signals cover every grant-making agency: repeat findings at the same grantee, and **auditor concentration** (one small audit firm auditing many grantees that share officers, or whose audits never report findings).

11.2 Audit-trail tests on internal ledgers (Phase 2)

These follow the journal-entry and fraud procedures in auditing standards (AU-C 240 and PCAOB AS 2401) and are run on the general ledger, vendor master, payment and card files that OIGs can obtain.

Test	Rule	Scheme
Sequence gaps and duplicates	Missing or repeated check, invoice and obligation numbers	Removed or fabricated documents
Three-way match	Payment without a matching order and receiving report, or with amounts that differ	Payment for goods not delivered
Vendor-employee match	Vendor address, phone or bank account equal to an employee's	Shell vendors owned by insiders
Bank account change	Vendor bank account changed shortly before a large payment, especially after an email request	Business email compromise, diversion
Segregation of duties	Same user created the vendor, entered the invoice and approved the payment	Insider fraud
Timing	Entries on weekends, holidays or after hours; entries just after period close	Concealment, manipulation
Just-below-approval	Entries clustered under a user's approval limit (bunching test, section 5.4)	Splitting to avoid review
Card tests	Split purchases (section 4.2), merchant category codes outside policy, purchases on leave days	Purchase and travel card misuse (Government Charge Card Abuse Prevention Act of 2012 requires OIG risk assessments of these programs)
Payroll	Employees with no leave ever, shared bank accounts, pay after separation	Ghost employees

12. Combining signals, calibrating and validating



12.1 Likelihood ratios

For a binary signal S, the case library gives

$$\mathrm{LR}^{+} = \frac{P(S \mid \text{known case})}{P(S \mid \text{population})},$$

estimated with additive smoothing so an unobserved combination never gets a ratio of zero. Under conditional independence, posterior odds are prior odds times the product of the signals' ratios:

$$\frac{P(F \mid S_1, \ldots, S_k)}{1 - P(F \mid S_1, \ldots, S_k)} = \frac{\pi}{1-\pi} \prod_j \mathrm{LR}_j.$$

Correlated signals (shared address and shared phone) are grouped and only the strongest in each group enters the product.

12.2 The base-rate trap

Even a strong signal leaves most flagged entities innocent when fraud is rare. A signal present in 40% of known cases and 0.5% of the population has $\mathrm{LR}^{+} = 80$; with a 1-in-1,000 prior, the posterior is only about 7%. This is why the methodology requires two independent signals and human review, and why scores are presented as rankings for review, not as probabilities of guilt. (The numbers in this paragraph are illustrative and are reproduced in the scoring tests.)

12.3 Validation metrics

- **Lift at the top q:** share of known cases in the top q of scores divided by their share overall. Release bar: lift of 5 or more at the top 5% (methodology section 5).
- **Precision at k:** share of known cases (or of hand-review "worth a look" judgments) among the top k leads, matched to investigator capacity.
- **Average precision:** area under the precision-recall curve; preferred over ROC AUC when cases are rare.
- **Calibration:** when scores are shown as probabilities in Phase 2, isotonic regression on held-out cases maps scores to observed rates.
- **Temporal holdout:** score the population as it looked before each case became public, so a screen is not credited for finding cases it learned from.

12.4 Controlling false discoveries

When the same test runs on m entities, rank their p-values $p_{(1)} \le \cdots \le p_{(m)}$ and flag the largest k with

$$p_{(k)} \le \frac{k}{m}\, q.$$



The Benjamini-Hochberg procedure keeps the expected share of chance hits among flagged entities at or below q (we use $q = 0.05$ for entity-level digit and outlier tests).

13. Methods by fraud area

Each fraud area in the roadmap maps to a small set of tests. "P1" means the test runs on public data in Phase 1; "P2" needs internal data.

Fraud area	Main schemes	Primary tests (section)	Data	Agencies (examples)
Health care billing	Upcoding, phantom services, kickback-driven referrals, hospice and home health mills	Peer outliers with shrinkage (5.3), concentration (5.6), growth (5.5), impossible days (4.3, P2), after-death services (4.4, P2), networks (8)	CMS provider and utilization files (P1); claims (P2)	HHS, VA, OPM (FEHB), DOL (FECA), USPS (workers' comp)
Procurement fronts	Pass-through set-aside fronts, shell bidders	Entity resolution (7), shared attributes and co-location (8.1-8.2), new entity with large award, subaward reconciliation (11.1)	SAM.gov, FPDS, USAspending subawards	DoD, VA, DOT, GSA, SBA, USACE, all agencies (U4)
Bid rigging	Rotation, cover bids, market allocation	Collusion screens and rotation test (6)	State DOT bid tabs; agency bid data (P2)	DOT, USACE, DoD, GSA
Price and cost mischarging	Inflated prices, defective pricing, labor mischarging	Unit price outliers (5.2), modification growth (U3), bunching under cost-data thresholds (5.4), impossible hours (4.3, P2)	FPDS, USAspending; invoices and timesheets (P2)	DoD, NASA, DOE, GSA
Grant misuse	Duplicate funding, unallowable costs, ghost participants	Text-similarity duplicate awards, Single Audit findings and auditor concentration (11.1), cost per participant outliers (5.2), drawdown gaps (11.1)	USAspending, fac.gov, Form 990, SBIR.gov	NSF, NIH, DOE, EPA, NEA, NEH, IMLS, regional commissions, LSC, CPB
Loans and guarantees	Identity and income fraud, lender and agent rings, straw borrowers	Entity resolution (7), broker centrality (8.3), digit tests on loan amounts (3), bunching under program caps (5.4)	SBA and USDA loan-level releases, EXIM authorizations, HMDA	SBA, USDA, HUD (FHA), EXIM, DFC



Fraud area	Main schemes	Primary tests (section)	Data	Agencies (examples)
Benefits	Ineligible, deceased or incarcerated recipients; identity reuse	Death, incarceration and identity tests (9), sampling estimates (10)	Aggregate public statistics (P1); payee files (P2)	SSA, RRB, DOL, VA, OPM, USDA
Tax credits	Preparer-driven false claims	Geographic outliers with shrinkage (5.3) on IRS statistics by ZIP; preparer networks (P2)	IRS SOI ZIP data (P1); returns (P2, section 6103 limits)	Treasury
Customs and trade	Undervaluation, false origin, transshipment	Mirror-trade gaps, importer entity resolution, co-location (8.2)	Census trade data, manifests	DHS
Subsidies (E-Rate, Lifeline, BEAD)	Consultant-steered bids, phantom subscribers	Collusion screens (6), broker centrality (8.3), duplicate subscribers (4.1, P2)	USAC open data	FCC, Commerce
Disaster and emergency aid	Duplicate claims, ineligible addresses	Duplicate tests (4.1), identity reuse (9), address validation	FEMA public data (P1); applicant files (P2)	DHS (FEMA), SBA, HUD (CDBG-DR)
Settlement and refund distributions	Synthetic claimants, duplicate claims	Duplicate and identity tests (4.1, 9) on claims files	Distribution reports (P1); claims (P2)	CFPB, FTC, SEC, CFTC, DOJ
Internal (every agency)	Card misuse, payroll ghosts, insider vendors, payment diversion	Split purchases (4.2), audit-trail tests (11.2), bunching (5.4)	OIG charge card risk assessments (P1); ledgers and card files (P2)	All bodies (U7)

14. Reference implementation

`analytics/forensic/` holds a small, readable implementation of every test in this handbook, with 30 unit tests that run on synthetic data:

Module	Sections
`digits.py`	3: Benford first, second, first-two and last-two digit tests; MAD conformity; per-digit z; summation; round numbers; duplication; relative size factor
`duplicates.py`	4: invoice normalization, four-tier duplicate test, split purchases, impossible days, after-death services
`outliers.py`	5: robust z, Poisson-Gamma shrinkage, CUSUM, bunching, Gini
`bids.py`	6: tender screens, rotation test, win-share matrix
`linkage.py`	7: Fellegi-Sunter EM, thresholds, Jaro-Winkler
`network.py`	8: shared-attribute projection, components, personalized PageRank, co-location



Module	Sections
`sampling.py`	10: stratified and ratio estimators, Neyman allocation, sample sizes, monetary unit sampling
`scoring.py`	12: likelihood ratios, posterior odds, lift, precision, average precision, two-signal rule

Run `python3 -m unittest discover -s analytics/tests -t analytics` from the repository root. Production pipelines may use faster libraries but must reproduce these functions' results on the test fixtures.

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- DOJ Procurement Collusion Strike Force

